

Automotive Sensors

Dynamic System Simulation

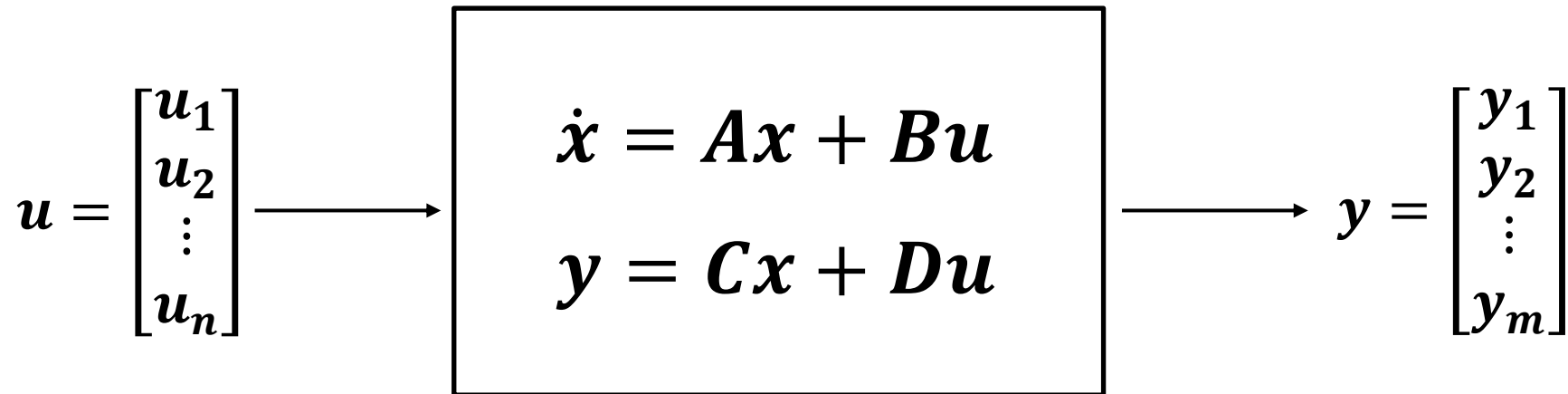
Automotive Intelligence Lab.

Linear System Modeling & Simulation

State Space Modeling of Dynamic System

■ Advantages of

- ▶ **Handles MIMO Systems Easily:** Naturally represents Multiple-Input Multiple-Output systems.
- ▶ **Explicit Internal State Variables:** Enables understanding and analysis of internal system behavior and variables.
- ▶ **Applicable to Various Analysis/Design Techniques:** Facilitates the use of modern control theories (stability, controllability, observability, optimal control, etc.).
- ▶ **Considers Initial Conditions:** Analyzes the impact of the system's initial state on its response.
- ▶ **Flexible Modeling of Nonlinear/Time-Varying Systems:** Wider range of system representation.
- ▶ **Easy for Simulation and Implementation:** Advantageous for computer simulations and digital controller design.



Examples of Continuous-Time Linear System

■ Mass-Spring-Damper system

- ▶ A classical second-order linear system where the force is linearly proportional to displacement and velocity.

$$m\ddot{x} = F - b\dot{x} - kx$$

$$\ddot{x} = \frac{F}{m} - \frac{b}{m}\dot{x} - \frac{k}{m}x$$

Let $x_1 = x, x_2 = \dot{x}, u = F$

Then

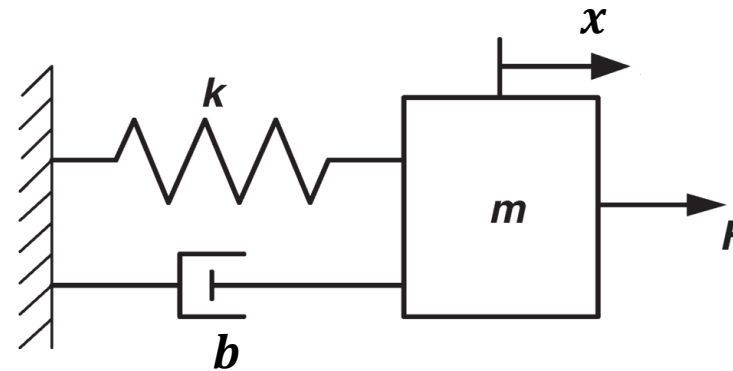
$$\dot{x}_1 = \dot{x} = x_2$$

$$\dot{x}_2 = \ddot{x} = \frac{u}{m} - \frac{b}{m}x_2 - \frac{k}{m}x_1$$

Matrix form

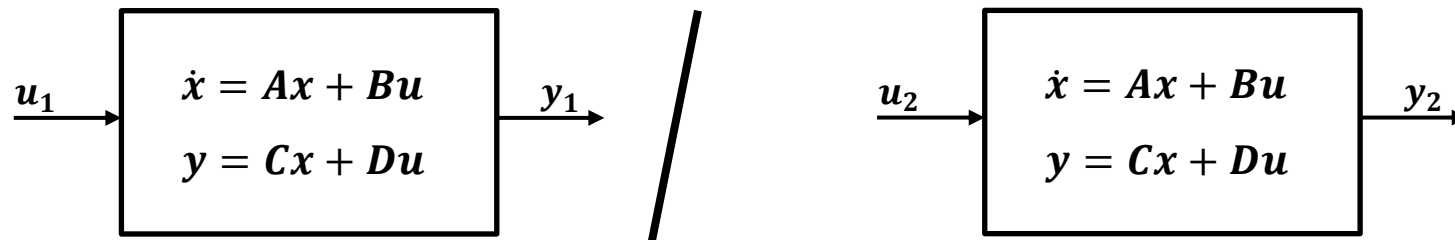
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow \text{measures } x(\text{position}) \text{ and } \dot{x}(\text{velocity})$$

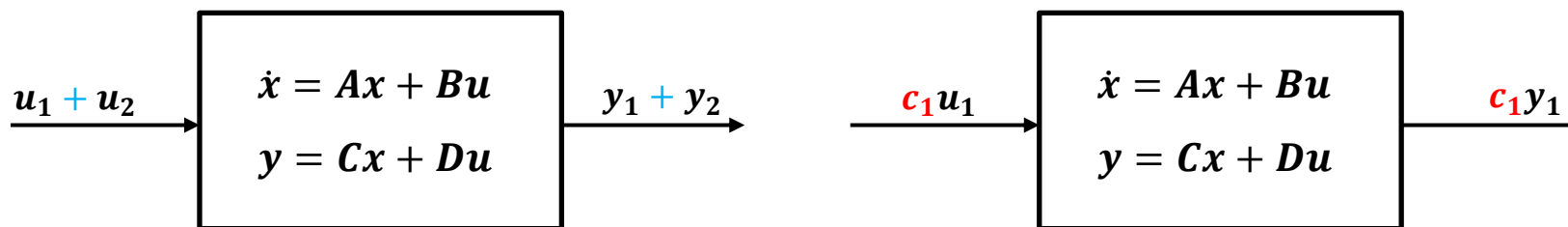


Linear System?

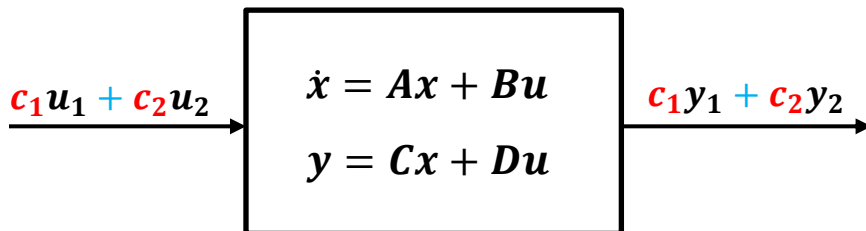
- u_1 input, y_1 output / u_2 input, y_2 output



- Additivity & Homogeneity



- Superposition Principle



Discrete-time Linear System

■ Equation

$$x_{t+1} = Fx_t + Gu_t$$

$$y_{t+1} = Hx_{t+1}$$

x : state

u : input

y : measurement

F : system matrix

G : input matrix

H : measurement matrix

■ Linear

- ▶ The change of the output is proportional to the change of the input.
- ▶ Can apply the **linear combination**
 - $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$

■ Linear model

- ▶ The **system matrix is constant** and independent of state.
- ▶ For any state, the change in the system is the same.

Examples of Discrete-time Linear System

- Final docking approach to the ISS can be approximated using a **Constant Velocity** or **Constant Acceleration** model
- **Constant Velocity(CV), Constant Acceleration(CA) model**
 - ▶ A motion model with linearly related position, velocity, and acceleration through time-discretized equations.



$$x_{t+1} = x_t + v_{x_t} dT$$

$$y_{t+1} = x_t + v_{y_t} dT$$

$$v_{x_{t+1}} = v_{x_t}$$

$$v_{y_{t+1}} = v_{y_t}$$

Equations of Constant Velocity Motion

$$x_{t+1} = x_t + v_{x_t} dT + \frac{1}{2} a_{x_t} dT^2$$

$$y_{t+1} = x_t + v_{y_t} dT + \frac{1}{2} a_{y_t} dT^2$$

$$v_{x_{t+1}} = v_{x_t} + a_{x_t} dT$$

$$v_{y_{t+1}} = v_{y_t} + a_{y_t} dT$$

Equations of Constant Acceleration Motion

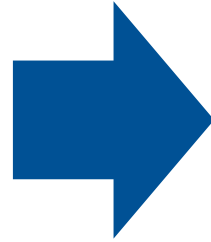
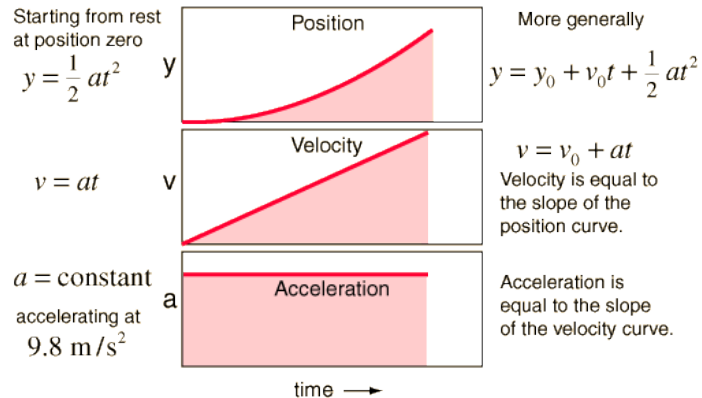
State Space Equation Modeling with CA

$$x_{t+1} = x_t + v_{x_t}dT + \frac{1}{2}a_{x_t}dT^2$$

$$y_{t+1} = x_t + v_{y_t}dT + \frac{1}{2}a_{y_t}dT^2$$

$$v_{x_{t+1}} = v_{x_t} + a_{x_t}dT$$

$$v_{y_{t+1}} = v_{y_t} + a_ydT$$



Linear state space equation

- ▶ $x_{t+1} = Fx_t + Gu_t$
- ▶ $y_{t+1} = Hx_{t+1}$

State and input definition

- ▶ $x_t = \begin{bmatrix} x_t \\ y_t \\ v_{x_t} \\ v_{y_t} \end{bmatrix}$, $u_t = \begin{bmatrix} a_{x_t} \\ a_{y_t} \end{bmatrix}$, $y_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$

System and control matrix from motion equation

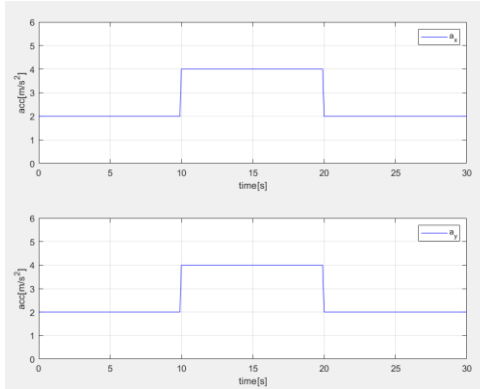
- ▶ $F = \begin{bmatrix} 1 & 0 & dT & 0 \\ 0 & 1 & 0 & dT \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $G = \begin{bmatrix} \frac{1}{2}dT^2 & 0 \\ 0 & \frac{1}{2}dT^2 \\ dT & 0 \\ 0 & dT \end{bmatrix}$

Measurement matrix

- ▶ $H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

CA Model Simulation Result

Input (a_x, a_y)



Acceleration

$$x_{t+1} = x_t + v_{x_t}dT + \frac{1}{2}a_{x_t}dT^2$$

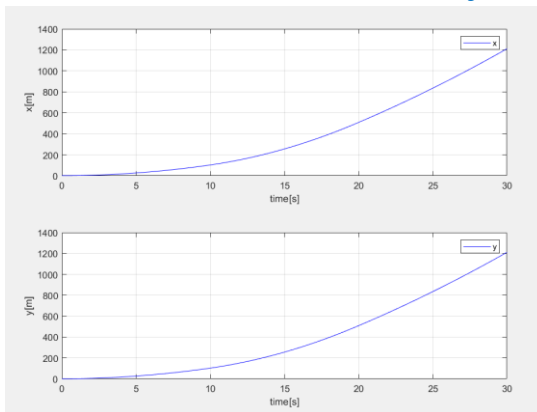
$$y_{t+1} = y_t + v_{y_t}dT + \frac{1}{2}a_{y_t}dT^2$$

$$v_{x_{t+1}} = v_{x_t} + a_{x_t}dT$$

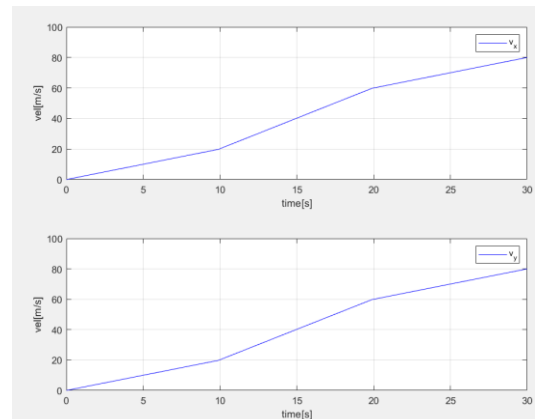
$$v_{y_{t+1}} = v_{y_t} + a_{y_t}dT$$

Constant Acceleration Model

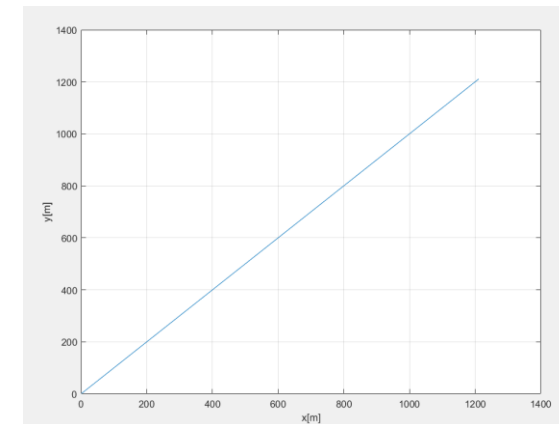
State (x, y, v_x, v_y)



Position



Velocity



Trajectory

Nonlinear System Modeling & Simulation

Discrete-time Nonlinear System

■ Equation

$$x_{t+1} = f(x_t, u_t)$$

$$y_{t+1} = h(x_t)$$

x : state

u : input

y : measurement

f : nonlinear function (system)

h : nonlinear function (measurement)

■ Nonlinear system

- ▶ The change of the output is **not proportional** to the change of the input.
- ▶ **Cannot** apply the linear combination.
 - $f(\alpha x + \beta y) \neq \alpha f(x) + \beta f(y)$
 - ex) $y = \sin(x)$, $y = x^2$, $y = \log(x)$
- ▶ Real system always have some nonlinearities.

■ Nonlinear model

- ▶ System matrix changes depending on state.
- ▶ In nonlinear models, the system matrix is a **function of state**.

Examples of Nonlinear System

■ Electronic circuits

- ▶ Semiconductor devices like diodes or transistors are nonlinear

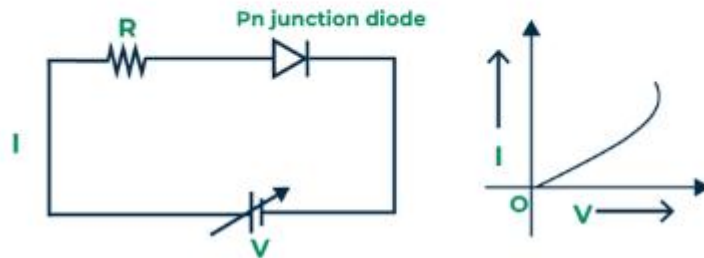
■ Nonlinear mechanical systems

- ▶ Mechanical systems can be nonlinear due to friction, inelastic impacts and large deformations of components

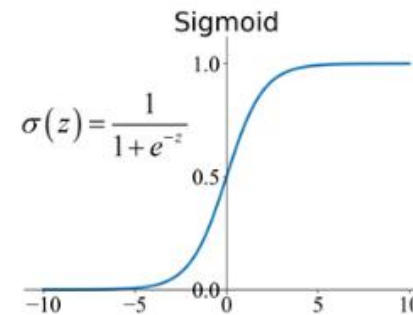
■ Artificial intelligence and neural networks

- ▶ Activation functions in neural networks are nonlinear functions

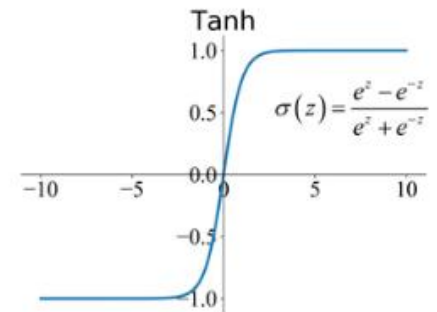
Non Linear Components



Example of nonlinear components



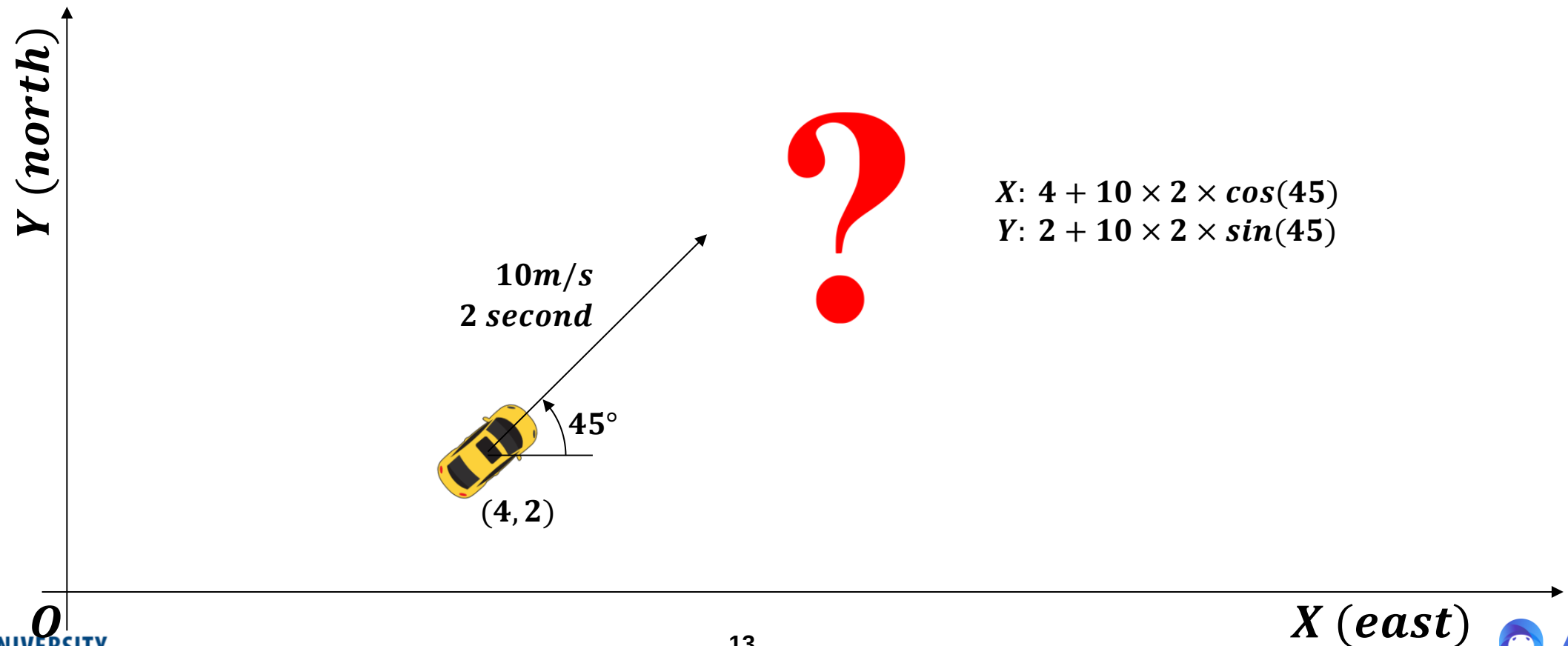
Example of activation functions



Dead Reckoning (I)

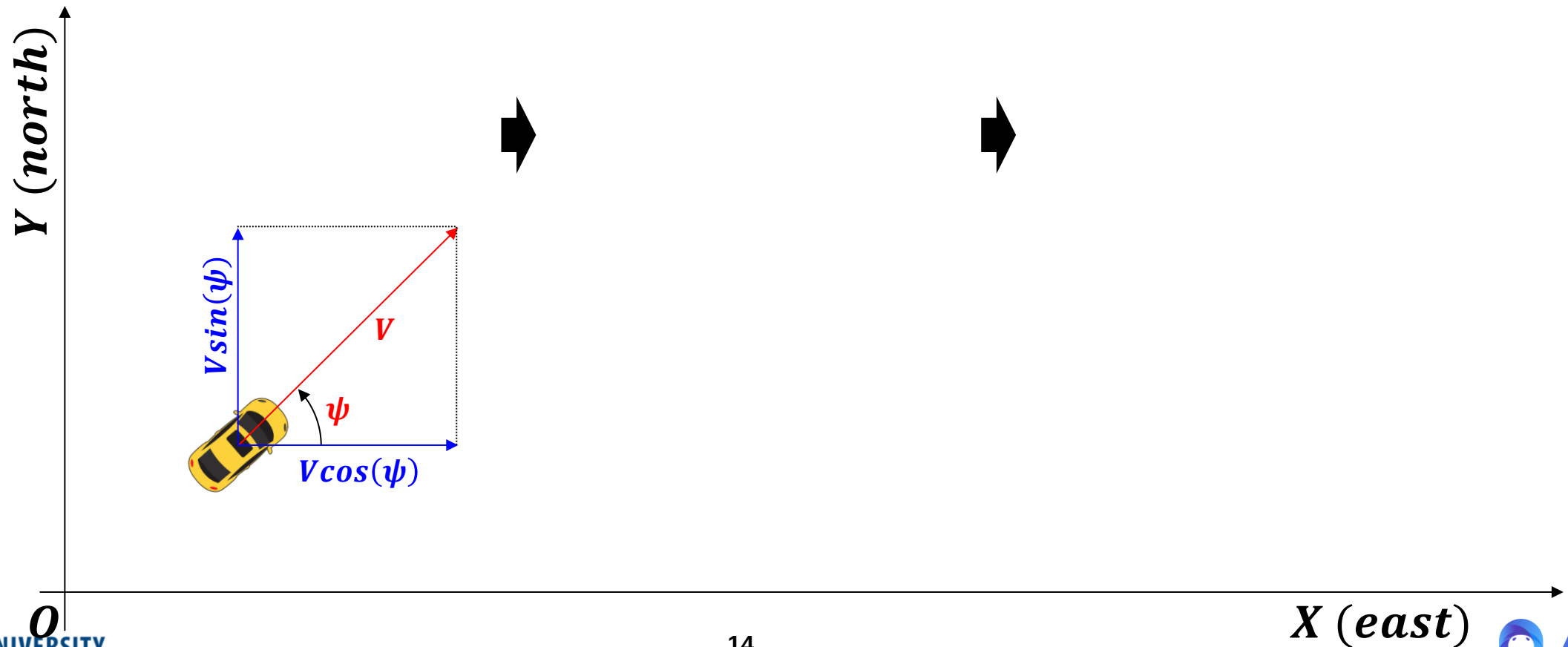
■ Dead reckoning is the process of

- ▶ calculating a current vehicle position by using
- ▶ 1) previously determined position
- ▶ 2) advancing that position based upon **speeds** over **elapsed time** and **course**.



Dead Reckoning (II)

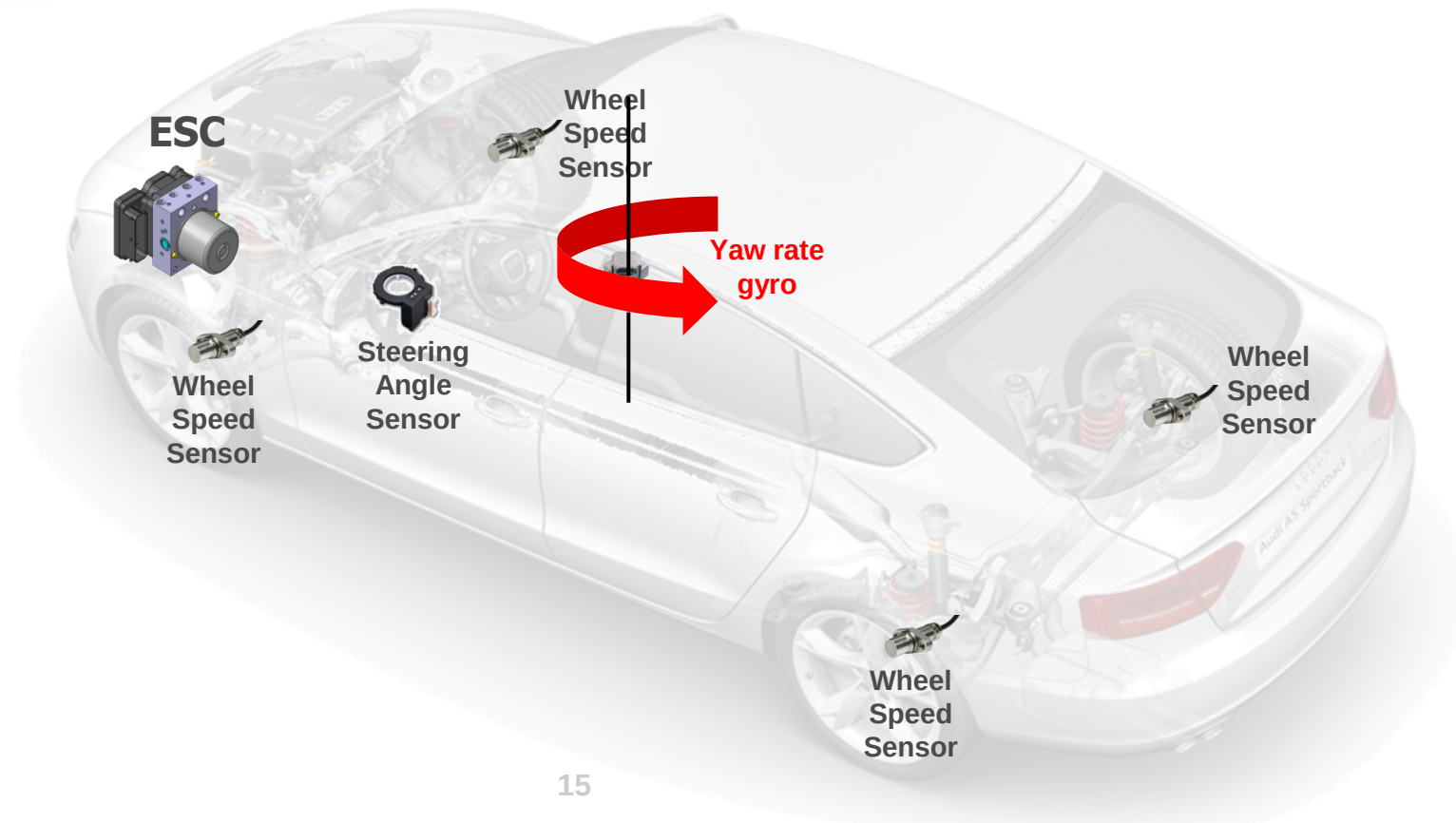
- Differential equation of dead reckoning when we have velocity V and yaw ψ
- Discretization using Euler method



Dead Reckoning using On-board sensor (I)

■ On-board sensors for Electronic Stability Control (ESC) system

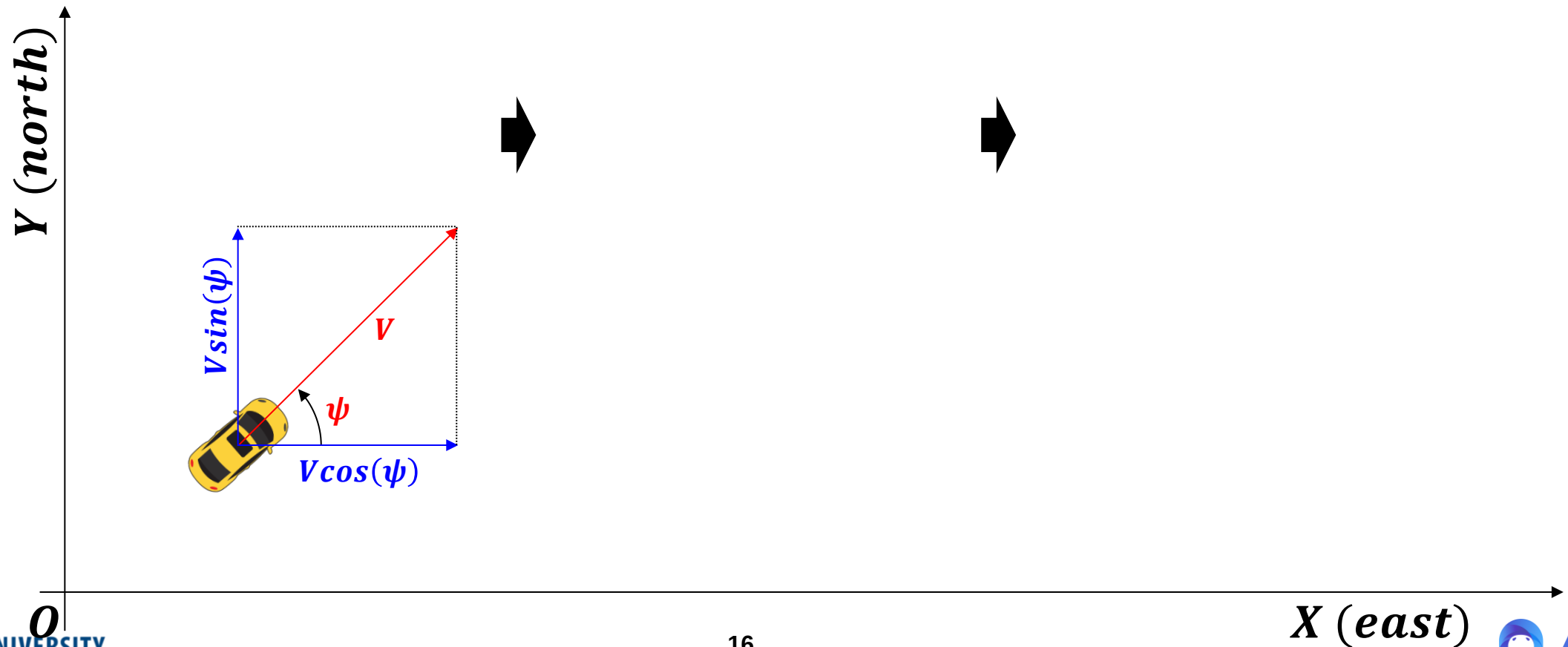
- ▶ Steering angle sensor
- ▶ Yaw rate gyro $\rightarrow \dot{\psi}_{gyro}$
- ▶ Wheel speed sensors $\rightarrow V_{wheel}$



Dead Reckoning using On-board sensor (II)

■ From the on-board sensor, we can obtain the

- ▶ Yaw rate ($\dot{\psi}_{gyro}$ deg/sec)
- ▶ Vehicle speed (V_{wheel} m/sec)



Dead Reckoning using On-board sensor (III)

- For on-board sensor input, V_{wheel} and $\dot{\psi}_{gyro}$

$$\begin{array}{lll}
 \dot{X} = V_{wheel} \cos(\psi) & \Rightarrow & \\
 \dot{Y} = V_{wheel} \sin(\psi) & \Rightarrow & \\
 \dot{\psi}_{gyro} \text{ is sensor value!} & \Rightarrow &
 \end{array}$$

Euler method

$$\dot{X} = \frac{X_{k+1} - X_k}{T}$$

$$\rightarrow X_{k+1} = X_k + T \cdot \dot{X}$$

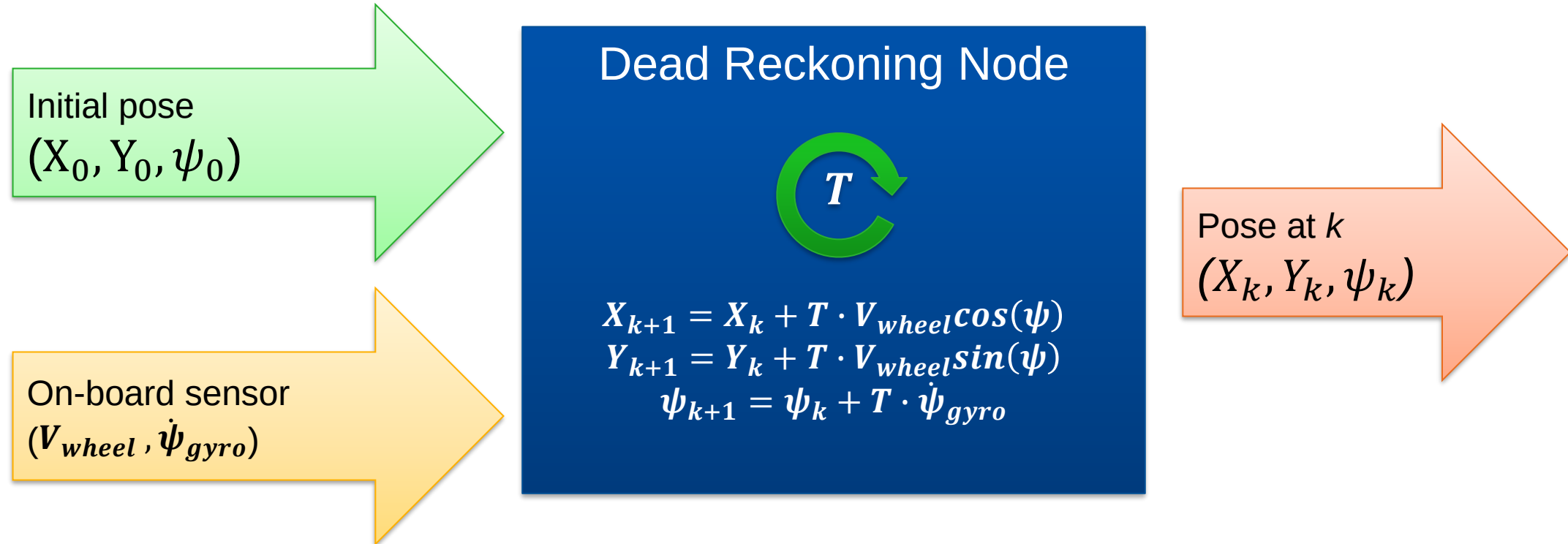
$$\begin{array}{ll}
 \Rightarrow X_{k+1} = X_k + T \cdot V_{wheel} \cos(\psi) \\
 \Rightarrow Y_{k+1} = Y_k + T \cdot V_{wheel} \sin(\psi) \\
 \Rightarrow \psi_{k+1} = \psi_k + T \cdot \dot{\psi}_{gyro}
 \end{array}$$

Dead reckoning using on-board sensor

$$\begin{aligned}
 X_{k+1} &= X_k + T \cdot V_{wheel} \cos(\psi_k) \\
 Y_{k+1} &= Y_k + T \cdot V_{wheel} \sin(\psi_k) \\
 \psi_{k+1} &= \psi_k + T \cdot \dot{\psi}_{gyro}
 \end{aligned}$$

Simulation of Dead Reckoning

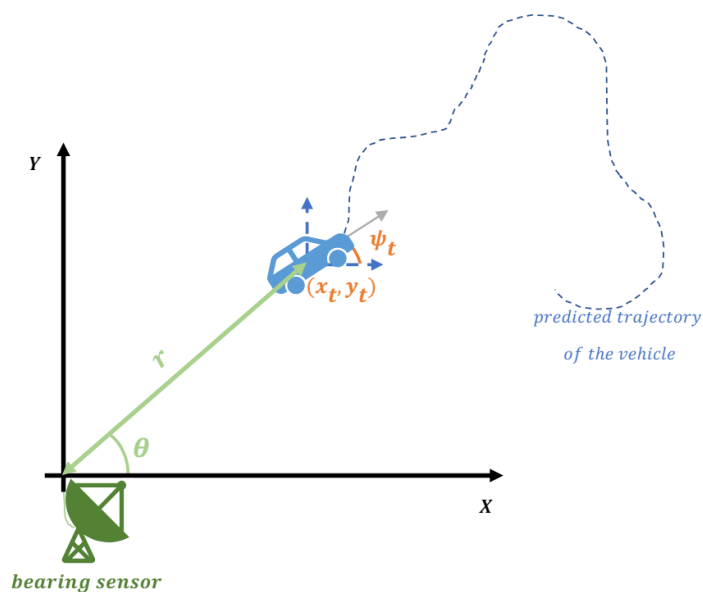
■ Simulation using ROS



DR with Longitudinal Acceleration Input

■ Vehicle model – Dead reckoning without speed sensor

- ▶ Estimates current position using speed, direction, and time from a known starting point
- ▶ Uses sensors like wheel encoders and gyroscopes; **no GPS needed**
- ▶ Simple and useful without GPS, but **errors build up over time** (drift)



$$x_{t+1} = x_t + \cos(\psi_t)v_t dT$$

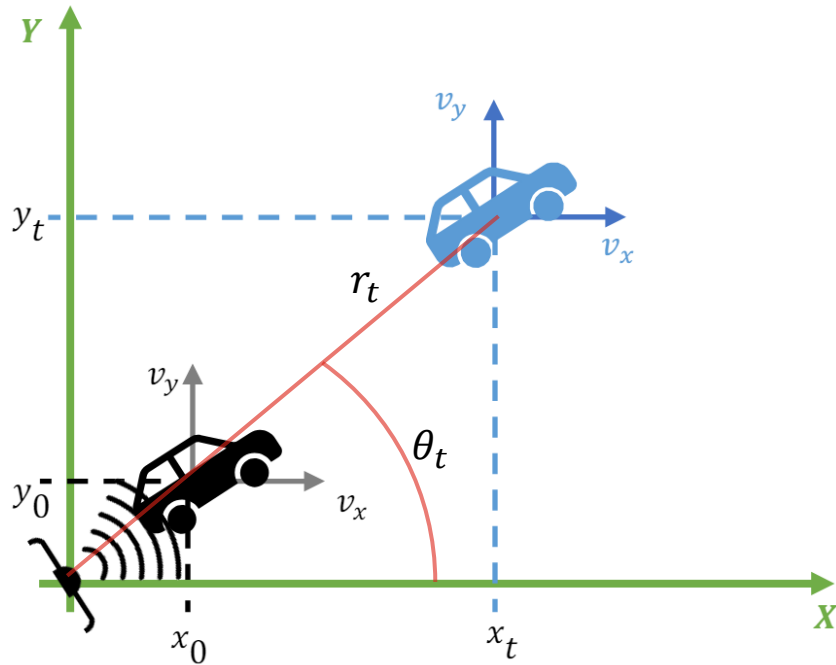
$$y_{t+1} = y_t + \sin(\psi_t)v_t dT$$

$$v_{v+1} = v_t + a_t dT$$

$$\psi_{t+1} = \psi_t + \omega_t dT$$

Equations of Dead Reckoning Motion

State Space Equation Modeling with DR



$$\begin{aligned}
 x_{t+1} &= x_t + \cos(\psi_t)v_t dT \\
 y_{t+1} &= y_t + \sin(\psi_t)v_t dT \\
 v_{t+1} &= v_t + a_t dT \\
 \psi_{t+1} &= \psi_t + \omega_t dT
 \end{aligned}$$

Equations of Dead Reckoning Motion

State Space Equation

- ▶ $x_{t+1} = f(x_t, u_t)$
- ▶ $y_{t+1} = h(x_t)$

State and input definition

- ▶ $x_t = \begin{bmatrix} x_t \\ y_t \\ v_t \\ \psi_t \end{bmatrix}, u_t = \begin{bmatrix} a_t \\ \omega_t \end{bmatrix}$

Vector representation of Dead Reckoning

- ▶ $\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ v_{t+1} \\ \psi_{t+1} \end{bmatrix} = \begin{bmatrix} x_t + \cos(\psi_t)v_t dT \\ y_t + \sin(\psi_t)v_t dT \\ v_t + a_t dT \\ \psi_t + \omega_t dT \end{bmatrix}$

Vector representation of measurement

- ▶ $\begin{bmatrix} r \\ \theta \end{bmatrix}_t = \begin{bmatrix} \sqrt{x_t^2 + y_t^2} \\ \tan^{-1}\left(\frac{y_t}{x_t}\right) \end{bmatrix}$



**THANK YOU
FOR YOUR ATTENTION**